Exam Review

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Coordinates

Let $B = \{e_1, \dots e_n\}$ be a basis of vector space V over the field $\mathbb{R}$. Then, the coordinates of vector $v \in V$ wrt. basis B are scalars $v_1, v_2, \dots v_n \in \mathbb{R}$ such that $v = v_1e_1 + v_2e_2 + \dots + v_ne_n$ is the unique linear combination of v in terms of B.

$$v \triangleq \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in {}^n\mathbb{R}$$

Transforming a linear combination into a matrix

If we can represent a vector using a column of scalars, then all the vectors in V where $\dim V = n$ can be cast as vectors in ${}^n\mathbb{R}$, and all the vectors in W where $\dim W = m$ can be remade as vectors in ${}^m\mathbb{R}$.

$L: V \to W$ can be represented by $m \times n$ matrix mapping ${}^n\mathbb{R}$ to ${}^m\mathbb{R}$.

$$\underbrace{w}_{\text{vector}} = L\underbrace{(v)}_{\text{vector}} = \sum_{j=1}^{n} v_j \underbrace{L(e_j)}_{\in W}$$

$$= \sum_{i=1}^{m} \underbrace{w_i}_{\text{scalar}} \underbrace{h_i}_{\text{vector in basis}}$$

$$\underbrace{w = L v}_{\text{scalars (coordinates)}}$$

Change of Basis

$$\underbrace{\begin{bmatrix} v_1^{(f)} \\ v_2^{(f)} \\ \vdots \\ v_n^{(f)} \end{bmatrix}}_{\substack{\text{coordinates of } v \\ \text{wrt. } B_f}} = \begin{bmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & \ddots & & \\ \vdots & & & \\ p_{n1} & & & p_{nn} \end{bmatrix} \underbrace{\begin{bmatrix} v_1^{(e)} \\ v_2^{(e)} \\ \vdots \\ v_n^{(e)} \end{bmatrix}}_{\substack{\text{coordinates of } v \\ \text{wrt. } B_e}}$$

$$v = \alpha_1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \qquad \text{coordinate: } \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

$$v = \beta_1 \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} + \beta_2 \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \qquad \text{coordinate: } \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

$$\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = P_{\alpha\beta} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_1 & \alpha_2 \\ -\alpha_2 & \alpha_1 \end{bmatrix} = \begin{bmatrix} \beta_1 + \beta_2 & \beta_1 - \beta_2 \\ -\beta_1 + \beta_2 & \beta_1 + \beta_2 \end{bmatrix}$$

$$\begin{cases} \alpha_1 = \beta_1 + \beta_2 \\ \alpha_2 = \beta_1 - \beta_2 \\ -\alpha_2 = -\beta_1 + \beta_2 \\ \alpha_1 = \beta_1 + \beta_2 \end{cases}$$

Determinants

Determinant function satisfies the following properties:

1. Multilinear in rows of A
2. Alternating in rows of A
3. Satisfies $f(I) = 1$

Note: If there are two equal rows, then the determinant is 0.

A function $f: {}^m\mathbb{R}^n \to \mathbb{R}$ is **multilinear** in the rows if for each $i = 1, 2, \dots m$, and for all $b, c \in \mathbb{R}$,

$$f\left(\begin{bmatrix} a_1 \\ \vdots \\ ba_i + ca_i' \\ \vdots \\ a_n \end{bmatrix}\right) = bf\left(\begin{bmatrix} a_1 \\ \vdots \\ a_i \\ \vdots \\ a_n \end{bmatrix}\right) + cf\left(\begin{bmatrix} a_1 \\ \vdots \\ a_i' \\ \vdots \\ a_n \end{bmatrix}\right)$$

It's a linear function of each of its rows while the other rows are held constant.

↳ multiplying any row by a scalar c changes the value of the determinant by a factor of c.

A function $f: {}^m\mathbb{R}^n \to \mathbb{R}$ is **alternating** on the rows if for all $i \neq k$ with $i, k \in \{1, 2, \dots m\}$,

$$f\left(\begin{bmatrix} a_1 \\ \vdots \\ a_i \\ \vdots \\ a_k \\ \vdots \\ a_n \end{bmatrix}\right) = -f\left(\begin{bmatrix} a_1 \\ \vdots \\ a_k \\ \vdots \\ a_i \\ \vdots \\ a_n \end{bmatrix}\right)$$

If you exchange two rows of A to form B, then $f(B) = -f(A)$.

↳ interchanging any two rows changes the sign of the determinant.

Property: Adding multiple of one row to another does not change determinant.

proof:

$$\det \begin{bmatrix} a_1 \\ \vdots \\ a_i + ca_j \\ \vdots \\ a_n \end{bmatrix} = \det \begin{bmatrix} a_1 \\ \vdots \\ a_i \\ \vdots \\ a_n \end{bmatrix} + \cancel{\underbrace{c \det}_{\text{multilinearity}} \begin{bmatrix} a_1 \\ \vdots \\ a_j \\ \vdots \\ a_n \end{bmatrix}}^{0} \leftarrow \text{two equal rows}$$

Property: Determinant of upper triangular matrix is product of diagonal entries.

Property: Determinant of non-invertible matrix is 0.

ij minor of matrix A is the $(n-1) \times (n-1)$ matrix formed by omitting the i^{th} row and j^{th} column of A. Denoted as A_{ij}.

(i,j)-cofactor of matrix $A \in {}^n\mathbb{R}^n$ is

$$C_{ij}(A) = (-1)^{i+j} \det M_{ij}(A)$$

$$\boxed{\det A = \sum_{j=1}^{n} a_{kj} C_{kj}}$$

Product Theorem: Let $A, B \in {}^n\mathbb{R}^n$. Then,

$$\boxed{\det AB = \det A \det B}$$

Adjoints

Let $A \in {}^n\mathbb{R}^n$. The adjoint of A, denoted $\operatorname{adj}A$, is the $n \times n$ matrix whose ij entry is $(-1)^{i+j} \det A_{ji}$

In cofactor form: $\boxed{\operatorname{adj}A = C^T}$, where $C = [C_{ij}] \in {}^n\mathbb{R}^n$ and $C_{ij}(A)$ is the (i,j)-cofactor of A.

Theorem: Let $A \in {}^n\mathbb{R}^n$. Then,

$$A \operatorname{adj}A = (\operatorname{adj}A)A = (\det A)I$$

If A is invertible, then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}A$$

Some fact from deepseek:

- If $\operatorname{rank} A < n-1$, then $\operatorname{adj} A = 0$
- If $\operatorname{rank} A = n-1$, then $\operatorname{rank}(\operatorname{adj} A) = 1$
- If $\operatorname{rank} A > n-1$, then $\operatorname{adj} A$ is invertible

Eigenproblem

$$Ap = \lambda p$$

$$(\underbrace{\lambda}_{\text{eigenvalue}} I - A)\underbrace{p}_{\text{eigenvector } (\neq 0)} = 0$$

$$\det(\lambda I - A) = 0$$

Definition: Let $A \in {}^n\mathbb{R}^n$. For a given eigenvalue λ, the **eigenspace** of A corresponding to eigenvalue λ is

$$\underbrace{E_\lambda(A)}_{\substack{\text{set of all vectors corresponding to} \\ \text{eigenvalue } \lambda \text{, together with the} \\ \text{zero vector}}} = \{x \in {}^n\mathbb{R} \mid Ax = \lambda x\} = \operatorname{null}(\lambda I - A)$$

Definition: Let $A \in {}^n\mathbb{R}^n$. The **characteristic polynomial** of A, $C_A(\lambda) \in P_n(\mathbb{R})$, is

$$C_A(\lambda) = \det(\lambda I - A)$$

Similarity

Let $A, B \in {}^n\mathbb{R}^n$. A and B are similar if there is an invertible $S \in {}^n\mathbb{R}^n$ such that $A = SBS^{-1}$.

Proposition: Let $A \in {}^n\mathbb{R}^n$ and $T \in {}^n\mathbb{R}^n$ be invertible. Then, characteristic polynomials of A and $T^{-1}AT$ are identical and so the eigenvalues of the two matrices are identical.

Two similar matrices have same characteristic polynomials and eigenvalues.

Diagonalization

$P \in {}^n\mathbb{R}^n$ diagonalizes $A \in {}^n\mathbb{R}^n$ if P is invertible such that

$$P^{-1}AP = \Lambda$$

$\Lambda = \operatorname{diag}\{\lambda_\alpha\}$ is a diagonal matrix of eigenvalues of A

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & \\ 0 & \dots & & \lambda_n \end{bmatrix}$$

Diagonalization theorem: $P \in {}^n\mathbb{R}^n$ diagonalizes $A \in {}^n\mathbb{R}^n$, where $P^{-1}AP = \Lambda$ if and only if the columns of P form a basis for ${}^n\mathbb{R}$ consisting of eigenvectors p_α of A, where $Ap_\alpha = \lambda_\alpha p_\alpha$

Corollary: $A \in {}^n\mathbb{R}^n$ is diagonalizable if and only if ${}^n\mathbb{R}$ has a basis consisting of eigenvectors of A.

Note: If A has complex eigenvalues then it is not diagonalizable as you would not have $\Lambda \in {}^n\mathbb{R}^n$.

Theorem: Let $A \in {}^n\mathbb{R}^n$ with distinct eigenvalues $\lambda_1, \dots \lambda_r$, $r \leq n$. If $x_a \in E_{\lambda_a} \setminus \{0\}$, then $\{x_1, x_2, \dots x_r\}$ is linearly independent.

Corollary: Let $A \in {}^n\mathbb{R}^n$ and suppose A has n distinct eigenvalues $\lambda_1, \dots \lambda_n$. Then A is diagonalizable.

Theorem: Let $A \in {}^n\mathbb{R}^n$ with distinct eigenvalues $\lambda_1, \dots \lambda_r$, $r \leq n$. If H_{λ_a} is a linearly independent set in E_{λ_a}, then

$$H = H_{\lambda_1} \cup H_{\lambda_2} \cup \dots \cup H_{\lambda_r} = \bigcup_{a=1}^{r} H_{\lambda_a}$$

is linearly independent and

$$m_1 + m_2 + \dots + m_r \leq n$$

where $m_a = \dim E_{\lambda_a}$.

Corollary: If $A \in {}^n\mathbb{R}^n$ has r distinct eigenvalues $\lambda_1, \dots \lambda_r$ and $m_a = \dim E_{\lambda_a}$, $a = 1, \dots r$ such that

$$m_1 + m_2 + \dots + m_r = n,$$

then A is diagonalizable.

Geometric and Algebraic Multiplicities

Let λ_i be an eigenvalue of $A \in {}^n\mathbb{R}^n$. If $C_A(\lambda) = (\lambda - \lambda_i)^{n_i} g(\lambda)$ where $g(\lambda) \neq 0$, then n_i is the **algebraic multiplicity** of λ_i. The **geometric multiplicity** is the dimension of the eigenspace $E_{\lambda_i}(A)$

Theorem: Suppose A has k distinct eigenvalues, $\lambda_1, \lambda_2, \dots \lambda_k$. If algebraic and geometric multiplicities for each eigenvalue of A are equal, then A is diagonalizable.

Diagonalization Test: Let $A \in {}^n\mathbb{R}^n$. If A is diagonalizable, then for each eigenvalue λ of A, the algebraic and geometric multiplicities of λ are equal.

contrapositive ↓

Theorem: If A has an eigenvalue whose algebraic and geometric multiplicities are not equal, then A is not diagonalizable.

Proposition: Let $A = \begin{bmatrix} I & B \\ 0 & C \end{bmatrix} \in {}^n\mathbb{R}^n$, where $B \in {}^r\mathbb{R}^{n-r}$, $C \in {}^{n-r}\mathbb{R}^{n-r}$, $I \in {}^r\mathbb{R}^r$ is the $r \times r$ identity matrix.

Then, $\det A = \det C$

$$A = \left[\begin{array}{cccc:cc} 1 & 0 & 0 & 0 & x & x \\ 0 & 1 & 0 & 0 & x & x \\ 0 & 0 & 1 & 0 & x & x \\ 0 & 0 & 0 & 1 & x & x \\ \hdashline 0 & 0 & 0 & 0 & x & x \\ 0 & 0 & 0 & 0 & x & x \end{array}\right] \quad C$$

Multiplicity theorem: Let λ_a be an eigenvalue of $A \in {}^n\mathbb{R}^n$. Then, $1 \leq m_a \leq n_a$, where m_a and n_a are respectively the geometric and algebraic multiplicities of λ_a. In particular, if $n_a = 1$, then $m_a = n_a = 1$.

First-order ODE's

$$\dot{x} = Ax$$

$$P^{-1}AP = \Lambda = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_r \end{bmatrix}$$

$$\Rightarrow x = Pn$$

$$\dot{n} = \Lambda n$$

$$\dot{n}_a = \lambda_a n_a, \quad a = 1, \dots n$$

$$\Rightarrow n_a(t) = ce^{\lambda_a(t)}$$

$$x(t) = Pn(t)$$

$$= P \begin{bmatrix} c_1e^{\lambda_1 t} \\ c_2e^{\lambda_2 t} \\ \vdots \\ c_ne^{\lambda_n t} \end{bmatrix}$$

$$= c_1e^{\lambda_1 t}p_1 + c_2e^{\lambda_2 t}p_2 + \dots + c_ne^{\lambda_n t}p_n$$

how to solve:

① Find eigenvalues and eigenspaces of A

If A diagonalizes, $A = P\Lambda P^{-1}$

② Build basis of solutions

③ Write down general solution

④ Solve IVP

Helpful manipulations

- $(PQ)^{-1} = Q^{-1}P^{-1}$
- $(SDS^{-1})^2 = SDS^{-1}SDS^{-1} = SD^2S^{-1}$
- $P^{-1}\lambda IP - P^{-1}DP = P^{-1}(\lambda I - D)P$
- $\lambda I = P\lambda IP^{-1}$
- $x_B = [T]_{B\alpha} x_\alpha$
- To prove injective: $\overset{\text{null } T}{\ker T} = \{0\}$, $\dim\ker T = 0$
- To prove surjective: $\dim \operatorname{im} T = \dim W$
- $\det S^{-1} = \frac{1}{\det S}$
- suppose $A = \begin{bmatrix} — a_1 — \\ \vdots \\ — a_n — \end{bmatrix}$, $B = \begin{bmatrix} | & & | \\ b_1 & \dots & b_n \\ | & & | \end{bmatrix}$

  first col of AB: Ab_1 $(\in \operatorname{col} A)$

  first row of AB: a_1B $(\in \operatorname{row} B)$

  $$\begin{bmatrix} — a_1 — \\ \vdots \\ — a_n — \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \operatorname{col} A$$

  $$\begin{bmatrix} x_1 & \dots & x_n \end{bmatrix} \begin{bmatrix} — a_1 — \\ \vdots \\ — a_n — \end{bmatrix} \in \operatorname{row} A$$
- $\dim \operatorname{null} A$ is the geo. mult of $\lambda = 0$

  $(\dim E_{\lambda=0} = \dim \operatorname{null} A)$
- sum of diagonal matrices is diagonal, add the entries
- non-invertible → $\det A = 0$
- diagonal matrix: product commutes $(\Lambda_A\Lambda_B = \Lambda_B\Lambda_A)$
- $\det A = \lambda_1\lambda_2 \dots \lambda_n$ (square matrix)
- $\operatorname{tr} A = \lambda_1 + \lambda_2 + \dots + \lambda_n$ (square matrix)
- similar matrices have same determinant
- eigenvalues of A^T same as A,

  eigenvalues of A^{-1} is reciprocal of A.